

# Sprint #1 (1/29 – 2/12)

Scrum Master: Jacob Sullins

## Project:

ETSU Marketplace is a university-only online marketplace that allows verified students to buy, sell, and exchange items and housing opportunities securely. It is a marketplace for students and only students to use. Built in ASP.NET MVC and C#.

## Activities

Building our project proposal presentation completed a lot of the “pre-processing” that Sprint 1 usually consists of. We already had a GitHub repository, Trello board, Discord server, and a set of planned product backlog items. Thus, many of the tasks for Sprint 1 were development or research related. As far as activities are concerned, they are primarily composed of the Scrum meetings during the class sessions and virtual discussions over Discord during the snow days. We also spent the first session of Sprint 1 adding more user stories to our product backlog, hoping to pre-fill it for future Sprint grooming. With our plan of product backlog items, we groomed them into tasks to place into our Sprint Backlog. Aside from planning Sprint 1 and Scrum meetings, we did not do many activities due to the weather and illness. We did hold a Sprint Retrospective virtually to close Sprint 1.

## Challenges

As mentioned previously, weather and illness did affect our first sprint, but nothing to make it too challenging. We found our way around the roadblocks preventing us from coming into class, talking remotely and ensuring everyone was on track to complete their tasks. Since the sprint was early in the semester, everyone’s workloads were not too heavy, allowing us to complete our sprint goal.

## Achievements

Our goal for this sprint was to have a general idea of how our project looks and how users can navigate through the website. By the end of the sprint, we had created a modern user interface using Bootstrap and Razor, basic implementations of features to build upon in future sprints, and an authentication system (which severely held my team back last semester). Due to everyone’s experience, we had achieved our sprint goal easily and quickly, giving us time to research tools to implement features that will be implemented later.

## Member Contributions & Owned Tasks

### Jabari Mitchell

Jabari started development first, adding some business logic that the foundation of our application will be built upon. He designed a parent class called Listing that ItemListing and LeaseListing inherit from. He also designed enums for ItemListing, these enums being Category and Condition. These classes will model the listings that users can view and create. He also created a homepage for our application, which was stylized to be accessible and attractive to users. This involved some use of Bootstrap and Razor, and he configured our HomeController to serve this page by default.

Owned tasks/user stories:

- As a student, I want to use a navigation bar to easily navigate to various areas on the website.
- As a student, I want to see a modern user interface so that the website is easy to browse.
- Create classes for Listing, User, and Item
- Create View for home page

### Palwasha Newell

Palwasha created pages for Leases and Items for sale, another pivotal part of the application. To implement this, several tasks had to be done. For reusability, we wanted Listings to be presented as cards that we could list many of on a screen when necessary. This was implemented as ViewModels to be passed into PartialViews. Partials act as reusable cshtml that display whatever ViewModel is passed into it. She created the ViewModel for a listing card and the partial that presents the listing card. She then created the Items and Leases pages which use the listing card partial. To achieve our goal of getting a general idea of what our application will look like, she created test data in the ListingsController to pass into the pages, simulating a populated marketplace. She also created some UI tests in Selenium to optimize her workflow; Selenium tested our navbar to ensure navigation was running smoothly.

Owned tasks/user stories:

- Create Listing cards for reusability
- Create ViewModel for Listing to pass into PartialView
- Create View for Items page
- Create View for Leases page
- Link buttons for Home and Shop

- Start development for Selenium UI testing

## Jacob Sullins

I handled basic authentication for this sprint, opting to use ASP.NET Identity for our application. Our application requires users to register and login with their ETSU emails before they can have access to the marketplace. An assumption for our application is that users will enter ETSU emails that exist. To implement this, I opted for ASP.NET Identity's default configuration. As a precursor, I set up a SQLite database and added a database context, the packages for ASP.NET Identity, and the services required for integration. I then created a dynamic partial view that renders based on login status. Then I used the ASP.NET code generation tool (an ASP.NET library provided by Microsoft) to create the logic and basic pages for authentication. I modified the generated code in two ways: I added regex to force user account emails to end in "@etsu.edu" and stylized the login and register pages. The result was a customized authentication system and account management stylized like the rest of the application.

Owned tasks/user stories:

- As a student, I want to login with my credentials so that I can access the Marketplace.
- As a student, I want to register using my university email address so that only verified students can access the marketplace.
- As a student, I want to log in to my account so that I can access marketplace features.

## Nathanael Wolfe

Nathanael spent the sprint searching for tools necessary for features to be implemented later. He researched the possibility of using ASP.NET Core SignalR to handle real-time communication between users, permitting our server to push messages via Web Sockets. He looked into how to set up hubs, manage user connections, and send messages to specific users or groups. He reviewed integration with SignalR and ASP.NET Identity so everything is secure. He also researched storing messages in the database using EntityFramework so conversations persist and explored basic scalability options in case the app needs to support more users in the future. We will be proceeding with this tool set for our application.

Owned tasks/user stories:

- Research C#/.NET compatible messaging library

## Team Velocity

We did not add story points to our tasks and user stories in this sprint. Therefore, there is no way to calculate our velocity. We will adjust our product backlog for the second sprint.

## Trello Board

| Card Name                                                                                                                                                       | Members | List Name       | Archived |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------------|----------|
| As a student, I want to see listings as cards on the home page so that I can browse items on the marketplace.                                                   |         | Product Backlog | FALSE    |
| As a student, I want to view all active listings so that I can browse items available for sale                                                                  |         | Product Backlog | FALSE    |
| As a student, I want to create a listing with a title, description, price, and images so that I can sell an item to other students.                             |         | Product Backlog | FALSE    |
| As a student, I want to edit my own listing so that i can update item information.                                                                              |         | Product Backlog | FALSE    |
| As a student, I want to delete my own listing so that I can remove items that are no longer available.                                                          |         | Product Backlog | FALSE    |
| As a student, I want to filter listings by category so that I can browse items more efficiently.                                                                |         | Product Backlog | FALSE    |
| As a student, I want to contact a seller through the platform so that I can ask questions about an item.                                                        |         | Product Backlog | FALSE    |
| As a student, I want to search for listings so that I can find more specific items.                                                                             |         | Product Backlog | FALSE    |
| As a student, I want to view a seller's profile to see their other listings.                                                                                    |         | Product Backlog | FALSE    |
| As a student, I want to view all active lease takeover requests so that I can browse potential homes for rent.                                                  |         | Product Backlog | FALSE    |
| As a student, I want to create a lease takeover requests with a title, description, monthly rent, and images so that I can have other students view my request. |         | Product Backlog | FALSE    |
| As a student, I want to edit my lease takeover requests so that I can update the information.                                                                   |         | Product Backlog | FALSE    |
| As a student, I want to delete my lease takeover request so that I can remove it when it is no longer available.                                                |         | Product Backlog | FALSE    |

|                                                                                                                               |                 |                 |       |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-------|
| As a student, I want to filter lease takeover requests by category so that I can browse homes more efficiently.               |                 | Product Backlog | FALSE |
| As a student, I want to contact a seller through the platform so that I can ask questions about their lease takeover request. |                 | Product Backlog | FALSE |
| As a student, I want to login with my credentials so that I can access the Marketplace.                                       | jacobsullins    | Sprint 1 - Done | TRUE  |
| As a student, I want to register using my university email address so that only verified students can access the marketplace. | jacobsullins    | Sprint 1 - Done | TRUE  |
| As a student, I want to log in to my account so that I can access marketplace features.                                       | jacobsullins    | Sprint 1 - Done | TRUE  |
| Create View for Items page                                                                                                    | palwashanewell  | Sprint 1 - Done | TRUE  |
| Create ViewModel for Listing to pass into PartialView                                                                         | palwashanewell  | Sprint 1 - Done | TRUE  |
| As a student, I want to see a modern user interface so that the website is easy to browse.                                    | jabarimitchell1 | Sprint 1 - Done | TRUE  |
| Research C#/.NET compatible messaging library                                                                                 | wolfen16        | Sprint 1 - Done | TRUE  |
| Create classes for Listing, User, Item                                                                                        | jabarimitchell1 | Sprint 1 - Done | TRUE  |
| Create View for Leases page                                                                                                   | palwashanewell  | Sprint 1 - Done | TRUE  |
| Create View for home page                                                                                                     | jabarimitchell1 | Sprint 1 - Done | TRUE  |
| As a student, I want to use a navigation bar to easily navigate to various areas on the website.                              | jabarimitchell1 | Sprint 1 - Done | TRUE  |
| Link buttons for Home and Shop                                                                                                | palwashanewell  | Sprint 1 - Done | TRUE  |
| Start development for Selenium UI testing                                                                                     | palwashanewell  | Sprint 1 - Done | TRUE  |
| Create Listing cards for reusability                                                                                          | palwashanewell  | Sprint 1 - Done | TRUE  |